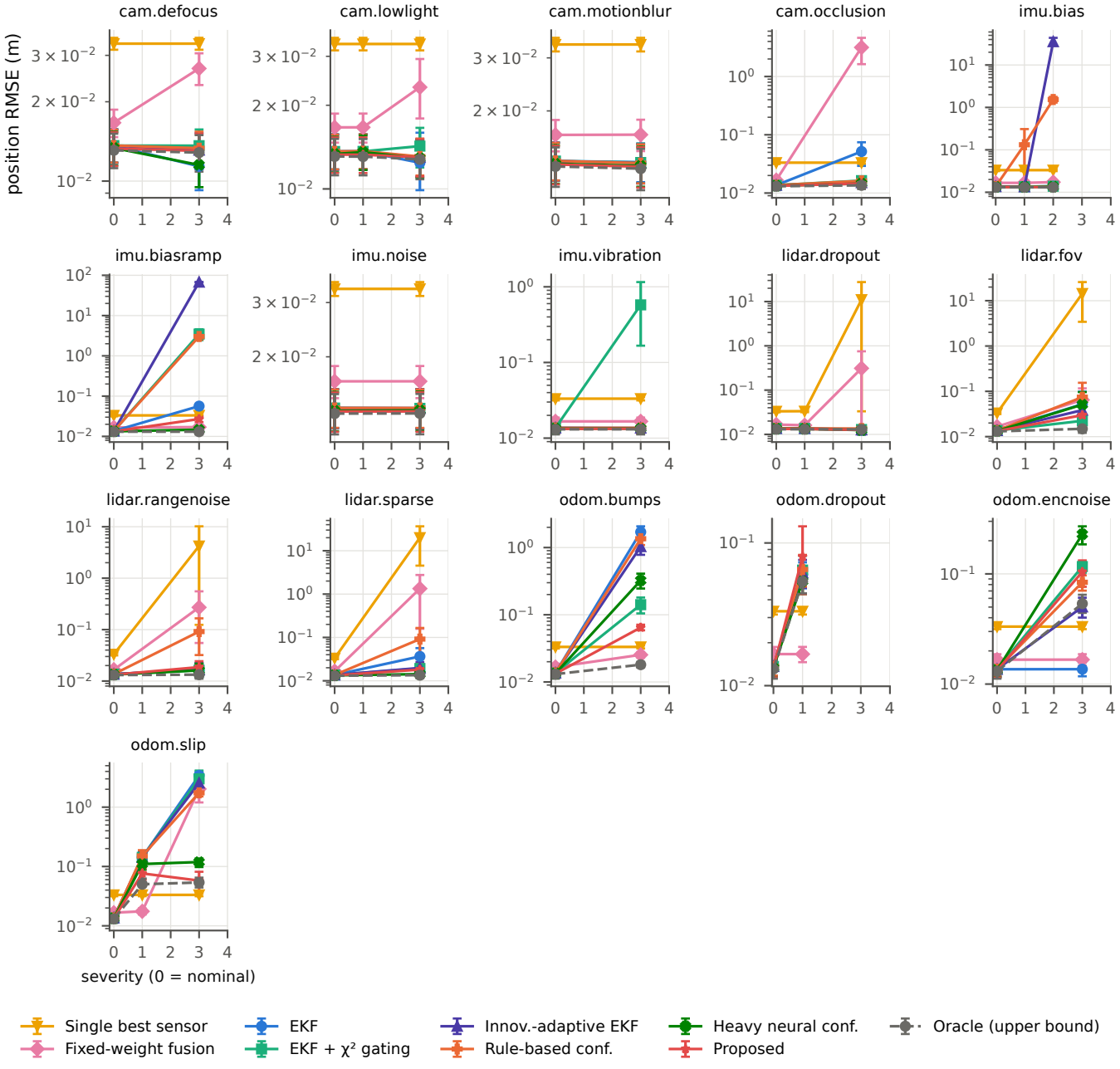

cam.defocus
cam.lowlight
cam.motionblur
cam.occlusion
imu.bias
imu.biasramp
imu.noise
imu.vibration
lidar.dropout
lidar.fov
lidar.rangenoise
lidar.sparse
odom.bumps
odom.dropout
odom.encnoise
odom.slip
position RMSE (m)
severity (0 = nominal)
Single best sensor
Fixed-weight fusion
EKF
EKF + χ² gating
Innov.-adaptive EKF
Rule-based conf.
Heavy neural conf.
Proposed
Oracle (upper bound)